

Image



Description

You might, for example, use the *Track* and the *Transporter* to model an AGV (automated guided vehicle) system. The distance, which the **Transporter** 🚚 has to travel on the *Track*—defined by the *Track*'s **Length**, the *Transporter*'s **MU Length**, and its **Speed**—determines the time, which the *Transporter* remains on the *Track*. As opposed to the point-oriented material flow objects, *Plant Simulation* uses the actual **Length**, which you type in, during your simulation run. A *Transporter* cannot pass another *Transporter* moving in front of it. *Transporters* thus retain their order of moving onto and of moving off the *Track* (FIFO)

If several *Transporters* travel along the *Track* at a different speed, and a faster one collides with a slower one, *Plant Simulation* activates the **Collision Control** of the faster *Transporter* and automatically reduces its speed to the speed of the slower *Transporter*.

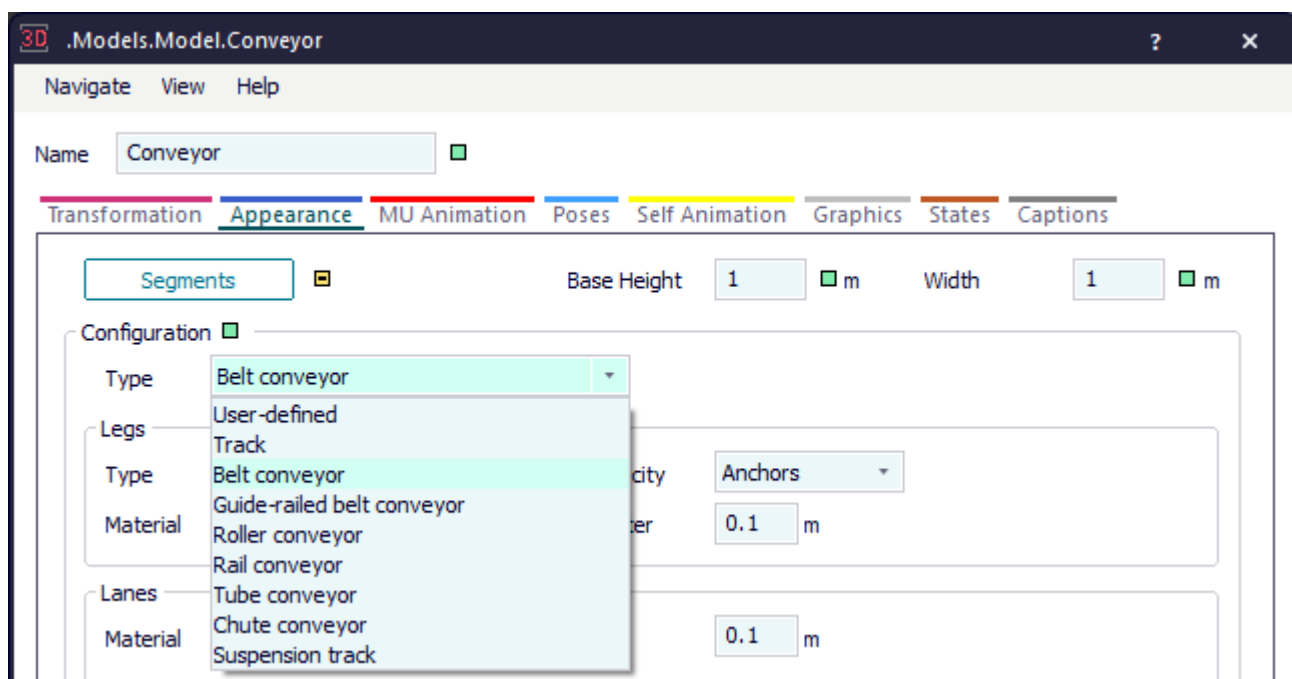
The maximum capacity of the *Track* is defined by its length and the lengths of the individual *Transporters* moving on it, i.e., a *Track* that is three yards long accepts three *Transporters* of one yard each at the most. You can use the **Capacity** to further restrict the number of *Transporters* located on the *Track*.

The *Transporter* can drive forward and **Backwards** on the *Track*, i.e., it drives onto the *Track* at its **Exit** and exits it at its **Entrance**. Driving forward and backward are not properties of the *Track*, but of the *Transporters*, which drive on it.

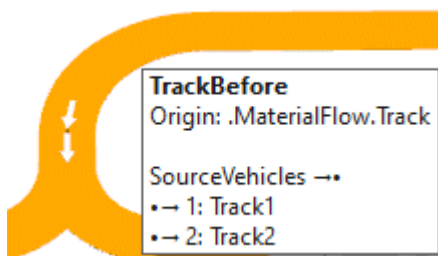
You can insert the *Track* into a *Frame* as follows:

- As a curved object, which is the default setting.
- By inserting any sequence of curved segments and straight segments, you can realistically model curved conveyor systems on which the *Transporters* move.

You can select different configurations for the *Track* on the tab **Appearance**.



To show a **tooltip** with information about the *Track*, hover with the mouse over it.

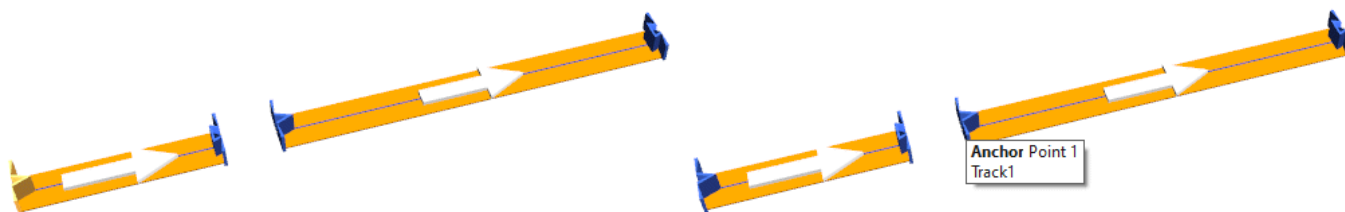


Show Manipulators

To change the length of the graphic and the anchor points of the object, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.



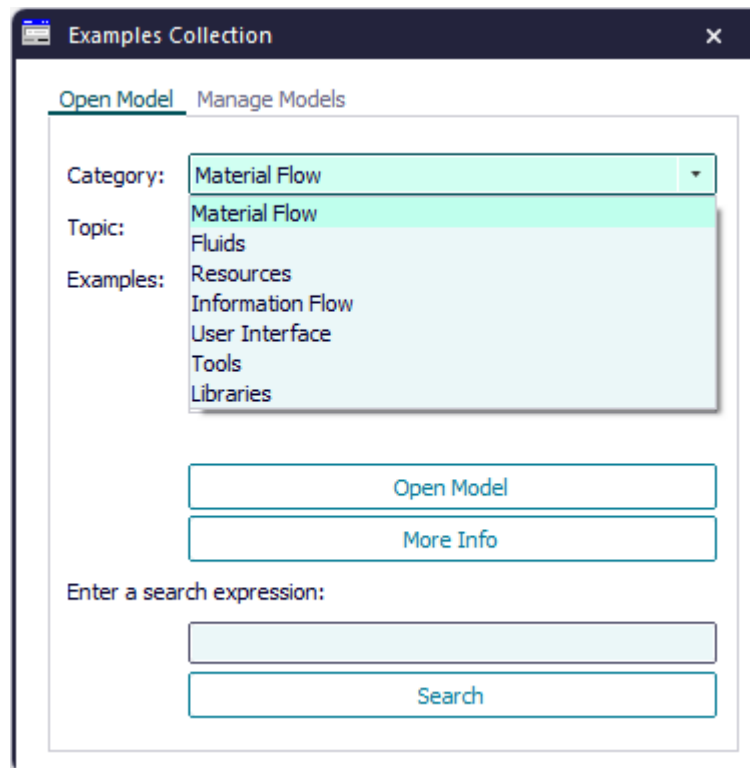
The manipulator at the start and at the end of the length-oriented object is cut off. When you attach an object of the same type to the other object, the two combined halves result in a complete manipulator again. When you move the mouse over a manipulator, *Plant Simulation* shows a *Tooltip*.



Add the Object to the Simulation Model

To add the object *Track* to your simulation model, click  **Manage Class Library** > **Basic Objects** > **MaterialFlow** > **Track** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page** > **Getting Started** > **Example Models** > **Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Model a Transport System with Passive Objects](#)

[Work with Length-oriented Objects](#)

[Routing \[TwoLaneTrack\]](#)

Routing [Track]

To facilitate a branching *Track*, can use any of these controls to move a *Transporter* on to the succeeding object.

Remarks

Plant Simulation uses:

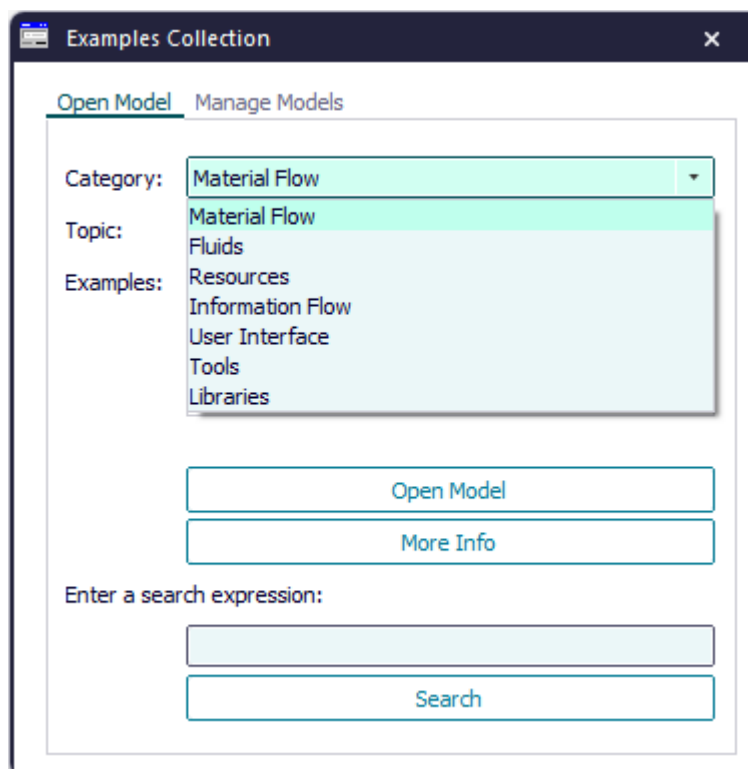
- The **Exit Control** of the *Track*.
- **Automatic Routing** of the *Transporter*.
- The **Driving Control** of the *Transporter*.
- The built-in properties of the *Track* described below.

The **Exit Control** of the *Track* has the highest priority. If you did not define an **Exit Control** and if you typed in a destination list for the *Transporter*, **Automatic Routing** moves the *Transporter* on to the proper successor, using the destination lists of the successors.

You can also assign a **Destination** to the *Transporter*. The *Track* has a **Forward Destination List** and a **Backward Destination List**, into which you type all destinations that the *Transporter* can reach when it drives on the *Track*. If you assigned a destination list to the *Transporter* that is ready to be moved on, the *Track* searches the **Forward Destination List** and the **Backward Destination List** of all **successors** for the **Destination** that you typed in, and, as soon as it finds it, moves the *Transporter* on to that successor. If the *Track* does not find the destination, it moves the *Transporter* to the next successor in turn. *Plant Simulation* terminates the search after it finds the first instance of the **Destination**. Automatic routing is inactive if you did not type in a **Destination** for the *Transporter* or if you typed in an **Exit Control** for the *Track*.

If you neither typed in an **Exit Control** nor any **Destination Lists** for the successors, the *Track* uses the **Driving Control** of the *Transporter* to move it. If you do not type in any controls at all, the built-in properties of *Track* move the *Transporters* to each of the successors, to which it is connected, in turn.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Model a Transport System with Passive Objects](#)

Dialog Box of the Track

Dialog Box of the Track

Double-click the icon of the *Track* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the [Tab Attributes](#).

Length [text box] - Track

Type the **Length** of the *Track* into the text box.

Length:  m

Remarks

After you inserted it, *Plant Simulation* shows its length here. The length of the *Track* and the sum of the lengths of all *Transporters* located on it determine how many *Transporters* it can accommodate at any one time.

You can change the length of the graphic and the anchor points of the *Track* by clicking [Show](#)

[Manipulators](#)  on the **Edit** ribbon tab or by pressing **M** on the keyboard.

SimTalk

Length [SimTalk] - Track

See also

OccupiedLength [SimTalk] - Track

Width [text box] - Track

Type the **Width** of the *Track* into the text box.

Width:  m

SimTalk

Width [SimTalk] - Track

Capacity [text box] - Track

Type in the **Capacity** of the *Track* into the text box.

Capacity: 

Remarks

The **Capacity** is the maximum number of *Transporters* that may be located on the *Track* as a whole or in part at any one time.

The default value **-1** stands for an infinite capacity.

SimTalk

Capacity [SimTalk] - Track

Backward Destination List [text box]

Click the ellipsis button and select a *List* object in the dialog **Select Object**. Or type the path to a list into the text box.

Backward Destination List: ... 

Remarks

Type all destination objects into this *List* object, which the *Track* can reach, when the *Transporter* moves in reverse on it.

Plant Simulation evaluates the destination lists when a *Transporter* is to be moved on to a successor. *Plant Simulation* searches for the destination of the *Transporter* in the destination lists of all succeeding *Tracks*, which are directly connected with *Connectors*. If a succeeding *Track* is connected via an *Interface*, this *Interface* should only have a single successor because *Plant Simulation* only searches in the first successor of the *Interface* for a destination list.

Note

The *Transporter* cannot only move forward but also in reverse on the *Track*, i.e., a **Transporter** can enter the *Track* at its exit and leave it at its entrance. Moving forward or in reverse are not properties of the object *Track*, but of the *Transporter* moving on it!

SimTalk

BwDestList [SimTalk]

See also

Select Object [for controls]

Forward Destination List [text box]

Type all destination objects into this *List* object, which the *Track* can reach, when the *Transporter* drives forward on it.

Forward Destination List: ... 

Remarks

Plant Simulation evaluates the destination lists when a *Transporter* is to be moved on to a successor. *Plant Simulation* searches for the destination of the *Transporter* in the destination lists of all succeeding *Tracks*, which are directly connected with *Connectors*. If a succeeding *Track* is connected via an *Interface*, this *Interface* should only have a single successor because *Plant Simulation* only searches in the first successor of the *Interface* for a destination list.

SimTalk

[FwDestList \[SimTalk\]](#)

See also

[Select Object \[for controls\]](#)

Tab Times

Define **Times** as described under the **Tab Times**.

Select a distribution from the drop-down list and type the values that this distribution requires into the text box. *Plant Simulation* shows the parameters along the upper border of the tab. You can also select a constant time (**Const**).

You can set the type of the distribution and a complete set of parameters with the method [setTypeAndAttr \[SimTalk\]](#).

See also

[Recovery Time \[general description\]](#)

[Recovery Time Starts](#)

Tab Failures

Define **failures** as described under the [Tab Failures](#).

Tab Controls [Track]

Provides controls to modify the built-in behavior of the object.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



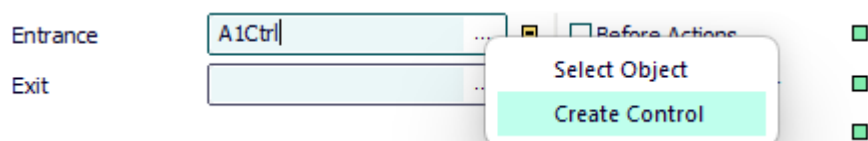
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

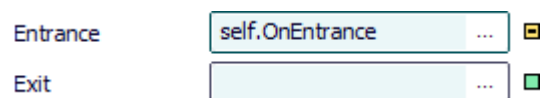
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.

- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

In addition, you can create and insert **Sensors**.

The *Track* shows the sensors, which you defined, on its graphic in the *Frame* window. To open the dialog **Sensors**, hold down **Alt** and double-click the sensor.

See also

Define Controls for Length-Oriented Objects

[Entrance Control \[general description\]](#)

[Exit Control \[general description\]](#)

[Backward Entrance Control](#)

[Backward Exit Control](#)

[Pull Control \[general description\]](#)

[Shift Calendar \[tab Controls\]](#)

Tab Exit

Select to which of its successors the object moves the *MU* on the [Tab Exit](#).

See also

[Blocking \[exit strategy\]](#)

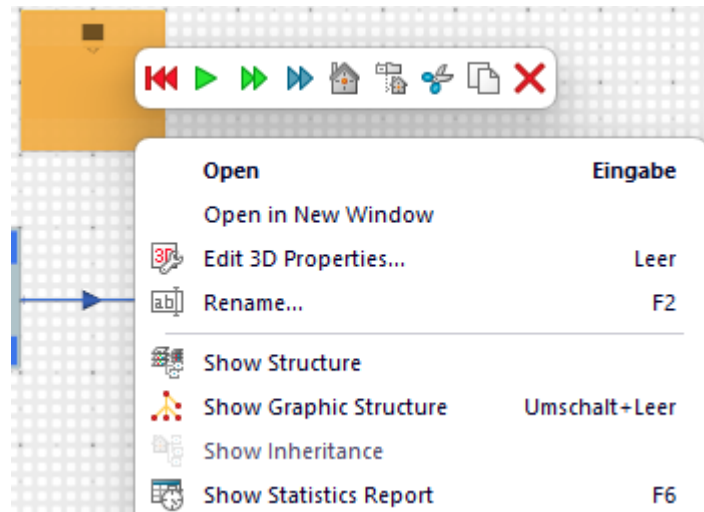
[Strategy \[material flow objects\]](#)

Tab Statistics

Statistics is described under the [Tab Statistics](#).

To view **Resource Statistics of Stationary Resources** in the **Statistics Report**, select **View > Show Statistics Report** in the dialog of the object.

You can also click the right mouse button in the *Frame* and select **Show Statistics Report** or you can press **F6**.



See also

[Resource Statistics \[check box\]](#)

[Resource Type](#)

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

The **View Menu** provides commands to access its functions.

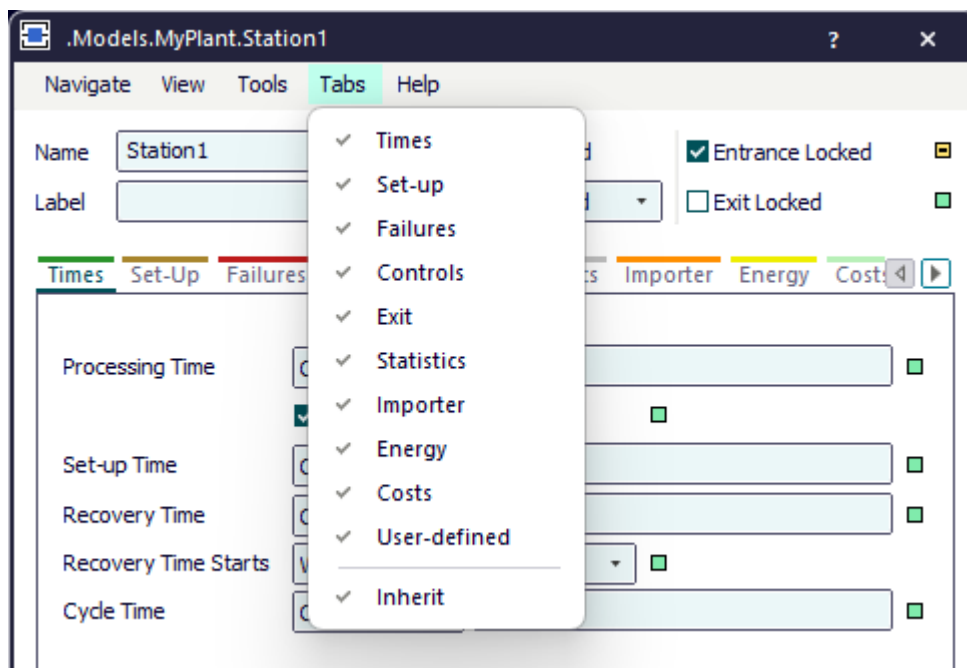
Refresh [on View menu]	Backward Blocking List
Show Statistics Report [on View menu]	Exit Blocking List
Show Attributes and Methods [on View menu]	Associated Lockout Zones
Contents [material flow objects]	Associated Shift Calendar
Forward Blocking List	

Tools Menu

The commands are described under the [Tools Menu](#).

Tabs Menu

Use the commands of the **Tabs** menu to show or hide individual tabs of the selected material flow objects. If you hide tabs that you do not need, *Plant Simulation* opens the dialog faster, and you can change to those tabs faster that you need in your daily work.



Remarks

To apply the changed settings, click **OK**, close the dialog, and reopen it.

The menu shows a check mark to the left of the displayed tabs.

The command **Inherit** turns inheritance of the displayed or hidden tabs in the dialog off or on.

Help Menu

The commands are described under the [Help Menu](#).

Methods of the Track

Methods of the Track

The *Track* provides:

- The *method* **getRouteLength [SimTalk] - Track.**
- The **_Methods of Curved Objects.**
- The **Methods of the Material Flow Objects.**
- The **Methods of All Objects.**

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.



Name	Value	Inherited	Watchable	Signature
contentsList	[]			([Contents:table]) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				([IconName:string, Width:integer, Height:integer]) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method

Identifier and data type of the parameter you enter

Data type of the return value



The screenshot shows the .Models.Model.Station window. The table lists the following methods:

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			([Index:integer:=1]) -> object
MUPart	VOID			([Index:integer:=1]) -> object

Below the table, the following labels are provided:

- Name of the method
- Identifier and data type of the parameter you enter
- Default value
- Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected **Class [general description]**.